

EnergyPlus-Trained Surrogate Models for Hourly Heating and Cooling Demand of the Vienna Building Stock

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Abstract

This study develops an EnergyPlus-trained surrogate for the building stock of Vienna. The stock comprises 134.6 million m² of gross floor area and is represented by eight residential and non-residential cohorts covering four construction periods. EnergyPlus is not applied to the city as a single large thermal zone. Instead, each cohort is simulated as a 1 000 m² reference archetype, and the resulting hourly useful heating and cooling intensities are scaled by the cohort floor areas obtained from Citiwatts. Two histogram gradient-boosting surrogate models reproduce the EnergyPlus outputs. Each combines an on/off classifier with a demand-magnitude regressor. Validation on complete calendar weeks excluded from training gives hourly R^2 values of 0.959 for heating and 0.975 for cooling. After floor-area-weighted aggregation, annual city demand differs from EnergyPlus by +1.7% for heating and -0.8% for cooling, while peak demand is underestimated by 6.7% and 10.1%, respectively. The complete EnergyPlus demand path for eight annual cohort simulations takes 67.1 s on the test system. Surrogate inference for the same 70,080 cohort-hours takes 1.73 s, corresponding to a speedup of approximately 39. The model provides fast EnergyPlus-consistent demand profiles; building flexibility and energy-system optimization are not considered.

Keywords: EnergyPlus; building stock; surrogate model; space heating; cooling; Vienna

1 Introduction

Hourly heating and cooling profiles are an important input to energy-system models with high shares of renewable energy. Annual demand alone does not show when heat pumps draw electricity, when heating coincides with a winter peak, or when cooling increases the summer load. These questions require a building model that responds to weather, solar and internal gains, occupancy schedules, the thermal envelope, and the operation of the heating and cooling systems. Simple degree-day relations or fixed normalized profiles can reproduce an annual total, but they do not represent these interactions explicitly.

EnergyPlus provides a considerably richer description of building demand [1, 2]. It solves the transient heat balance and accounts for heat transfer through opaque surfaces and windows, ventilation and infiltration, solar radiation, internal gains, thermal storage,

and HVAC controls. Heating and cooling loads therefore follow from the same physical balance instead of being imposed as independent profiles. EnergyPlus is also a widely used and extensively documented building-performance simulation program. This does not make every EnergyPlus result accurate by default: geometry, material properties, schedules, and weather still have to represent the case under study. Given suitable inputs, however, EnergyPlus offers a more defensible basis for deriving hourly building loads than a profile assembled from annual energy and outdoor temperature alone.

The detail that makes EnergyPlus useful also complicates its integration into larger models. A city-scale application requires several representative buildings, input files and schedules, repeated annual simulations, and the extraction of results from EnergyPlus output files. Direct coupling becomes particularly burdensome when building demand is evaluated repeatedly in scenario analyses or optimization models. In such settings, the building simulation can become a computational and technical bottleneck even when an individual annual run is not prohibitively slow.

A surrogate model offers a practical division of work. EnergyPlus is retained offline as the detailed teacher, while a statistical model learns the mapping from weather, operating conditions, and building properties to useful heating and cooling demand. Once trained, the surrogate can be called by an energy-system model without constructing and executing a new EnergyPlus run [3, 4]. The gain in speed is only useful if annual energy, hourly variation, seasonal transitions, and peak demand remain close to the teacher simulation.

This paper demonstrates this approach for the Vienna building stock. Citiwatts floor area and construction-period data are represented by eight residential and non-residential cohorts. EnergyPlus simulations for the 2023 weather year provide hourly useful heating and cooling, which are emulated by a gradient-boosting surrogate. The analysis addresses three questions: how accurately the surrogate reproduces held-out hourly EnergyPlus results; whether this accuracy is retained after aggregation to the Vienna stock, including seasonal and peak periods; and how much computation time is saved. Building flexibility, dispatch, and energy-system optimization are outside the scope.

2 Materials and Methods

2.1 Case Study and Building-Stock Representation

The case study covers the building stock within the Vienna boundary represented in the Citiwatts inventory [5]. The inventory reports a gross floor area of 134.6 million m² and a heated volume of 403.7 million m³. Residential buildings account for 89.0 million m² and non-residential buildings for 45.5 million m². Reported annual heat consumption is 17.54 TW h/a. This value includes domestic hot water and serves as a top-down energy-balance reference; it is distinct from the useful space-heating output of the EnergyPlus simulations.

The stock was divided by sector and construction period. Residential and non-residential buildings were each assigned to four periods: before 1975, 1975–1990, 1990–2000, and 2000–2014. The corresponding raw shares were 54%, 24%, 4%, and 17%, respectively, and were normalized before application within each sector. The resulting eight cohorts sum to the complete Citiwatts floor area. They represent an aggregation of the Vienna stock rather than a sample of individual buildings.

Table 1 reports the floor-area weights and period-dependent envelope parameters used in EnergyPlus. The U-value classes are informed by Austrian TABULA/EPISCOPE apartment-building data [6, 7]. In the absence of a corresponding Vienna non-residential typology, the same construction-period classes were used for both sectors. Residential solar transmittance values are proxies based on comparable TABULA window classes; the non-residential models use a uniform value of 0.60.

Sector-specific geometry was retained outside the period classification. For residential buildings, wall, window, roof, and exposed-floor areas were 0.82, 0.18, 0.37, and 0.36 m² per m² conditioned floor area, respectively. The corresponding non-residential ratios were 0.30, 0.12, 0.30, and 0.30. Residential areal heat capacities decreased from 80 to 65 W h/(m² K) across the four periods; non-residential values decreased from 70 to 55 W h/(m² K). The residential geometry is TABULA-informed, whereas the non-residential geometry and all heat-capacity values are explicit model assumptions.

Residential domestic-hot-water demand is estimated in the wider Vienna data set at 15.5 kW h/(m² a) and separated from the inventory heat total. Domestic hot water is not simulated or predicted in this study. Similarly, the annual Citiwatts heat consumption is not a regression target. Citiwatts defines the stock extent and provides the floor-area weights used for city aggregation. A subsequent energy-system application may normalize the EnergyPlus-derived profile and apply an external annual energy total, but this operation is not part of the validation below.

2.2 EnergyPlus Reference Simulations

One EnergyPlus 26.1 model was prepared for each cohort [2]. The models represent equivalent stock archetypes rather than particular buildings. A conditioned reference floor area of 1 000 m² was used in all simulations.

This common reference area keeps the geometry at a plausible archetype scale and permits the output to be expressed as demand intensity. It does not restrict the represented stock to 1 000 m². Envelope areas, internal gains, and air volumes were scaled consistently with the reference area, so the resulting specific demand can be applied to the corresponding cohort floor area.

The Citiwatts volume-to-area ratio gives a reference room height of approximately 3 m and a conditioned volume of approximately 3 000 m³. Each archetype was simulated for the 8760 hours of 2023. Outdoor temperature and solar radiation were obtained from Open-Meteo and converted to a pseudo-EPW weather file [8]. A Climate.OneBuilding file for Wien-Innere-Stadt supplied EPW header and format information but not the hourly observations [9]. Heating and cooling setpoints were 22 °C and 27 °C. The simulations included transmission through the envelope, solar and internal gains, ventilation, infiltration, and ideal-load heating and cooling. EnergyPlus returned hourly useful space-heating and useful-cooling demand. The separate winter experiment shown in Fig. 1 illustrates heat-balance differences between residential construction periods and was not added to the training data.

The outputs were converted to specific hourly demand $q_{c,t}^{\text{EP}}$ in kW h/(m² h). City-level demand was obtained by multiplying this intensity by the represented floor area A_c and summing across cohorts:

$$Q_{c,t} = q_{c,t}^{\text{EP}} A_c, \quad Q_t^{\text{Vienna}} = \sum_{c=1}^8 Q_{c,t}. \quad (1)$$

Equation (1) represents the complete inventory area through eight specific demand profiles. It is not a calibration of the EnergyPlus annual sum to the 17.54 TW h Citiwatts heat balance, which includes end uses outside the EnergyPlus targets. The unnormalized EnergyPlus series is retained so that teacher and surrogate are compared on the same basis.

2.3 Dataset Generation and Preprocessing

Combining eight cohort simulations with 8760 time steps produced 70,080 observations. One observation describes one cohort during one hour. Two target variables were derived: useful space-heating demand and useful cooling demand, both in kW h/(m² h). Separate models were trained for the two targets.

The explanatory variables cover four parts of the EnergyPlus input:

- *weather*: outdoor-air temperature, global horizontal irradiation, direct normal irradiation, and diffuse horizontal irradiation;
- *operation*: heating and cooling setpoints, internal gains, infiltration, and ventilation air-change rates;
- *time*: sine and cosine transformations of hour of day and day of year; and
- *building*: sector, cohort indicators, envelope U-values, envelope-area ratios, and areal heat capacity.

Table 1: Vienna stock weights and period-dependent envelope parameters. The U-values are common to both sectors. Residential g -values are cross-TABULA proxies; the uniform non-residential value is a model assumption.

Period	Represented floor area [million m ²]		U-value [W/(m ² K)]				Solar transmittance	
	A_{res}	A_{nonres}	U_{wall}	U_{win}	U_{roof}	U_{floor}	g_{res}	g_{nonres}
< 1975	48.6	24.8	1.40	2.30	1.70	1.20	0.85	0.60
1975–1990	21.6	11.0	1.10	2.70	0.80	0.80	0.76	0.60
1990–2000	3.6	1.8	0.60	2.50	0.50	0.50	0.60	0.60
2000–2014	15.3	7.8	0.35	1.40	0.20	0.40	0.50	0.60
Total	89.0	45.5						

The periodic time transformations avoid artificial discontinuities between hour 23 and hour 0 and between the end and beginning of the year.

A specific heat-loss coefficient was calculated from the products of U-values and their corresponding envelope-area ratios. Heating and cooling temperature differences were calculated between outdoor temperature and the respective setpoint. Their products with the heat-loss coefficient were included as additional variables. These engineered terms expose the principal thermal driving forces to the tree ensemble without using an EnergyPlus result as an input.

All cohort-hour combinations were retained; no random subsampling was applied. Data preparation required eight complete annual profiles, checked for duplicate cohort–timestamp pairs, and rejected missing values. Targets were normalized by represented cohort area before fitting. Floor-area values were used again only after prediction to obtain the city-level series.

2.4 Surrogate Model

Heating and cooling were modelled independently using the histogram-based gradient-boosting implementation in scikit-learn [10–12]. Gradient boosting constructs an additive ensemble of decision trees in which each iteration reduces the residual error of the current ensemble. This model class accommodates the combination of continuous weather and envelope parameters, cyclic time variables, and categorical cohort information used here.

Both targets contain long periods without demand. A single regression model gave excessive influence to these hours and produced residual summer heating. A two-part hurdle formulation was therefore used for each service. A `HistGradientBoostingClassifier` first estimates the probability $\hat{p}_t$ that demand is present. A `HistGradientBoostingRegressor` estimates the demand magnitude $\hat{q}_t^{\text{reg}}$. During inference, the non-negative regression output is retained if the probability exceeds a service-specific threshold τ :

$$\hat{q}_t = \begin{cases} \max(0, \hat{q}_t^{\text{reg}}), & \hat{p}_t \geq \tau, \\ 0, & \hat{p}_t < \tau. \end{cases} \quad (2)$$

Monotonic constraints were applied where the expected physical direction is unambiguous. Predicted heating cannot increase with outdoor temperature and

increases with the heating temperature difference. Cooling follows the opposite temperature response. Other variables remain unconstrained.

The regression component used a learning rate of 0.05, up to 700 boosting iterations, at most 159 leaf nodes per iteration, and an L2 regularization coefficient of 0.01. The classifier used up to 300 boosting iterations, 63 leaf nodes, and an L2 coefficient of 0.05. These settings define the active v2 surrogate.

2.5 Model Training and Validation

All 70,080 observations entered the cross-validation procedure. Sample weights prevented moderate- and zero-demand hours from dominating the regression loss. Positive observations up to the median positive demand received weight 2, whereas observations in the highest 15% of positive demand received weight 9. Remaining observations, including zero-demand hours, received weight 1. For the classification component, positive and zero classes were balanced within each training fold.

The hurdle threshold was selected independently in each training fold from 0.10 to 0.90 in increments of 0.05. The selection criterion retained energy during true on-hours while penalizing demand during physically implausible off-hours. These were hours with outdoor temperature at or above the heating setpoint for heating, and at or below the heating setpoint for cooling. Threshold selection used training-fold predictions only. The selected threshold was 0.10 in all heating folds and ranged from 0.65 to 0.70 in the cooling folds.

Temporal generalization was assessed by four-fold grouped cross-validation, using the ISO calendar week as the grouping variable. All hours of a given week, across all eight cohorts, were assigned to the same fold. Adjacent hours were therefore not randomly divided between training and validation data, and each reported prediction refers to a week excluded from the corresponding fit. All cohorts were represented in each fold. The procedure evaluates unseen periods within the 2023 weather year; it does not test a previously unseen building class or an independent climate year.

Held-out cohort-hour predictions were evaluated with the coefficient of determination (R^2), mean absolute error (MAE), and root mean squared error. Predictions were subsequently scaled according to Eq. (1) to obtain a fully out-of-fold Vienna time series. Annual relative error, peak-hour relative error, and Pearson correlation

Table 2: Four-fold calendar-week cross-validation of the active surrogate.

Metric	Heating	Cooling
Hourly R^2	0.959	0.975
MAE [kW h/(m ² h)]	5.8×10^{-4}	5.4×10^{-4}
City annual error [%]	+1.7	−0.8
City peak-hour error [%]	−6.7	−10.1
City Pearson r	0.982	0.989

were calculated for that series. The deployable models were finally refitted to the complete annual data set; these full-data fits were not used to report accuracy.

2.6 Runtime Assessment

Computational performance was measured for all eight annual cohort profiles on one computer. The EnergyPlus path included schedule and IDF generation, execution of the simulation engine, and extraction of hourly results from the SQL output. Engine time was measured three times per cohort and aggregated from the cohort medians. Inference for both surrogate targets was repeated five times and the median duration was reported. Plot generation was disabled. The one-time training duration was recorded separately and excluded from the inference speedup.

3 Results

3.1 EnergyPlus Cohort Response

The EnergyPlus heat balances differ markedly between residential construction periods (Fig. 1). During the selected winter day, approximate transmission losses decline from about 82 W/m² for the pre-1975 cohort to about 26 W/m² for the 2000–2014 cohort. Ventilation losses are similar because the same air-change assumption is used across cohorts. Infiltration is omitted from the figure because its common rate does not distinguish construction periods.

3.2 Surrogate Accuracy

Table 2 summarizes the out-of-fold accuracy. At cohort-hour level, the surrogate reaches an R^2 of 0.959 for heating and 0.975 for cooling. After floor-area-weighted aggregation, annual city demand differs from EnergyPlus by +1.7% for heating and −0.8% for cooling. The temporal correlations of the city series are 0.982 and 0.989. The largest discrepancy occurs at the annual maximum: peak heating is underestimated by 6.7% and peak cooling by 10.1%.

Annual heating errors for the large residential cohorts remain within a few percent. The largest cohort-relative error occurs for non-residential buildings from 2000–2014 (+23%). Their EnergyPlus heating intensity is only 3.8 kW h/(m² a); consequently, a small absolute deviation produces a large percentage error and has little influence on the aggregate Vienna result. Annual

Table 3: Runtime for eight cohorts and 8760 hours.

Path	Prep. [s]	Engine [s]	SQL [s]	Total [s]
EnergyPlus	27.5	24.8	14.8	67.1
Surrogate inference	—	—	—	1.73

cooling errors remain within approximately $\pm 2\%$ for all cohorts.

The selected seasonal weeks in Fig. 2 show how the aggregate error is distributed in time. During the peak-heating week, the surrogate follows the daily variation but underestimates the highest loads; total heating energy over the week is 1.6% below EnergyPlus. In the early-April heating week, energy is 3.5% higher and the peak magnitude is closely reproduced. Cooling profiles in late August and September show close agreement in most hours. The absolute August cooling peak remains underestimated, consistent with the annual peak metric.

Figure 3 relates city demand to outdoor temperature. The 1-K bin means produced by the surrogate follow those of EnergyPlus for both services. Heating decreases as outdoor temperature rises, whereas cooling increases at high temperatures. The transition between the two regimes is retained without a substantial band in which both services are simultaneously overpredicted.

EnergyPlus produces 0.43 GW h of heating from June through August, compared with 1.84 GW h from the surrogate. This residual is small relative to the approximately 5.3 TW h annual EnergyPlus space-heating series, but it indicates that the on/off transition is not reproduced perfectly.

3.3 Computational Performance

The complete EnergyPlus demand path takes 67.1 s, including 27.5 s for input preparation, 24.8 s for the simulation engine, and 14.8 s for SQL extraction (Table 3). Surrogate inference for both targets takes 1.73 s, corresponding to a speedup of 38.8. Engine wall-clock time alone is 14.3 times the inference time. The one-time fit of both surrogate models requires 18.5 s.

The inference comparison assumes that the hourly input table is available; loading and preparing a new weather table are not included. For repeated evaluations, 50 city years would require approximately 3 355 s on the EnergyPlus path and 87 s for surrogate inference, a difference of about 55 min.

4 Discussion

The results show that a compact statistical model can retain most of the hourly variation generated by the EnergyPlus cohort simulations. Annual energy and temporal correlation are reproduced more accurately than the maximum load. This distinction matters for energy-system applications. The surrogate is suitable for producing hourly demand shapes that are subsequently normalized to an independently specified annual energy balance. It should not be used without qualification

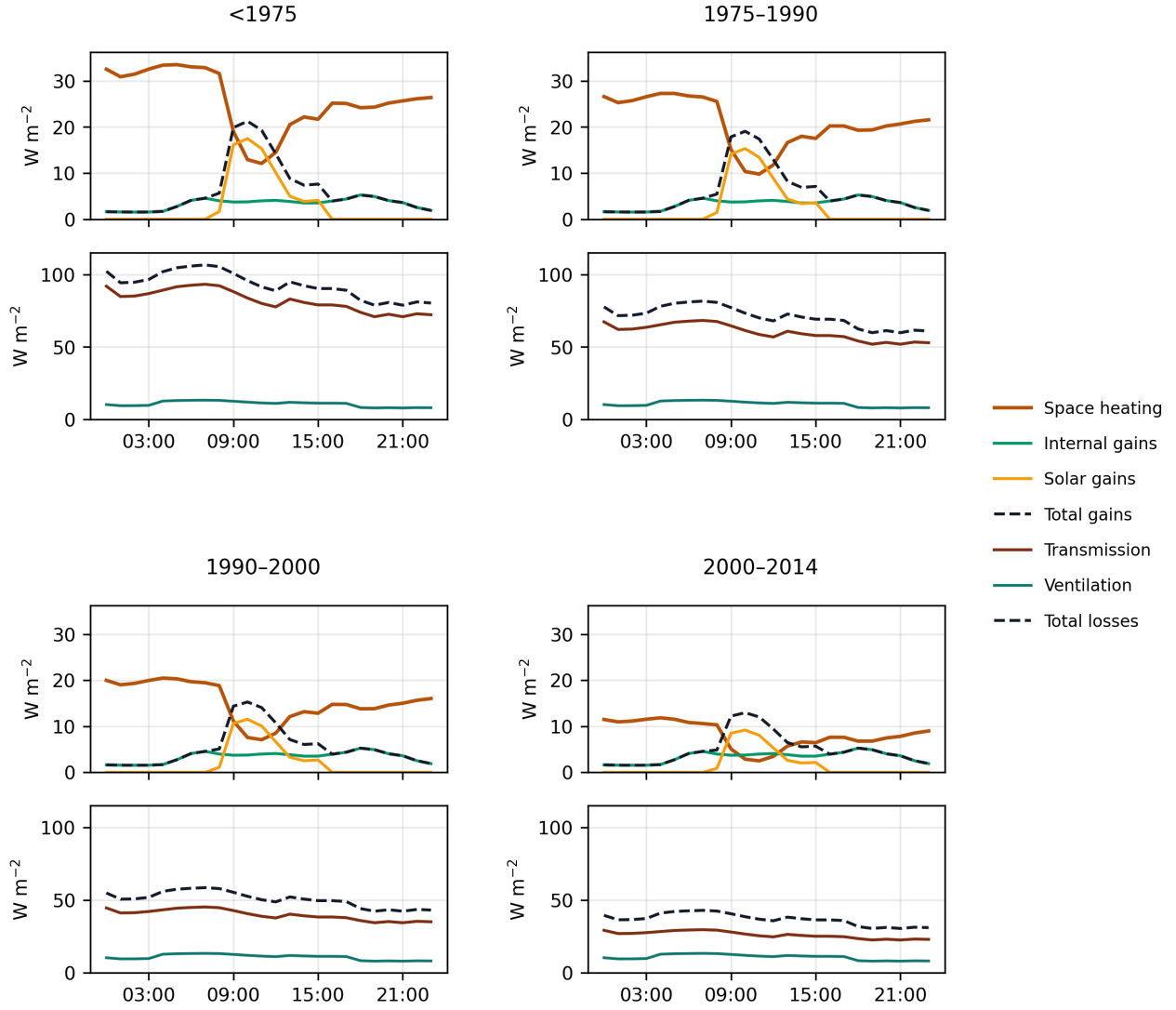


Figure 1: EnergyPlus teacher heat-balance flows for four residential construction periods during a winter day. Upper panels show space heating, internal gains, solar gains, and total gains. Lower panels show approximate transmission and ventilation losses. Common axis limits permit direct comparison between construction periods.

when the EnergyPlus maximum itself is treated as a design load.

The city-level results describe all floor area in the Citiwatts inventory, but they do not resolve every building in Vienna. Buildings within a given sector and construction period share one equivalent archetype and one operating profile. Aggregation therefore retains major differences in envelope quality while suppressing variations in geometry, occupancy, refurbishment state, and control behaviour within each cohort. The non-residential archetypes carry additional uncertainty because Vienna-specific geometry and window data were not available at the same level as for Austrian residential apartment buildings.

The validation is deliberately separated in time: complete calendar weeks are excluded from each fit. It is nevertheless limited to one weather year and to the same eight cohorts used during training. The reported accuracy therefore supports interpolation across unseen periods of 2023, but not transfer to a different climate, an extreme year outside the training range, or an unseen

building type. The remaining underestimation of peak loads is consistent with this limitation.

Finally, the surrogate reproduces reference heating and cooling demand only. It does not contain set-point changes, preheating, load curtailment, rebound, or indoor-temperature state transitions. These processes require a separate stateful response model before building flexibility can be represented in an optimization problem.

5 Conclusions

An EnergyPlus-to-surrogate workflow was developed for hourly useful heating and cooling demand of the Vienna building stock. Eight archetypes represent the complete Citiwatts floor area across two sectors and four construction periods. Their EnergyPlus outputs form a 70,080-row annual data set. Separate histogram gradient-boosting hurdle models reproduce held-out cohort-hour demand with R^2 values of 0.959 for heating and 0.975 for cooling.

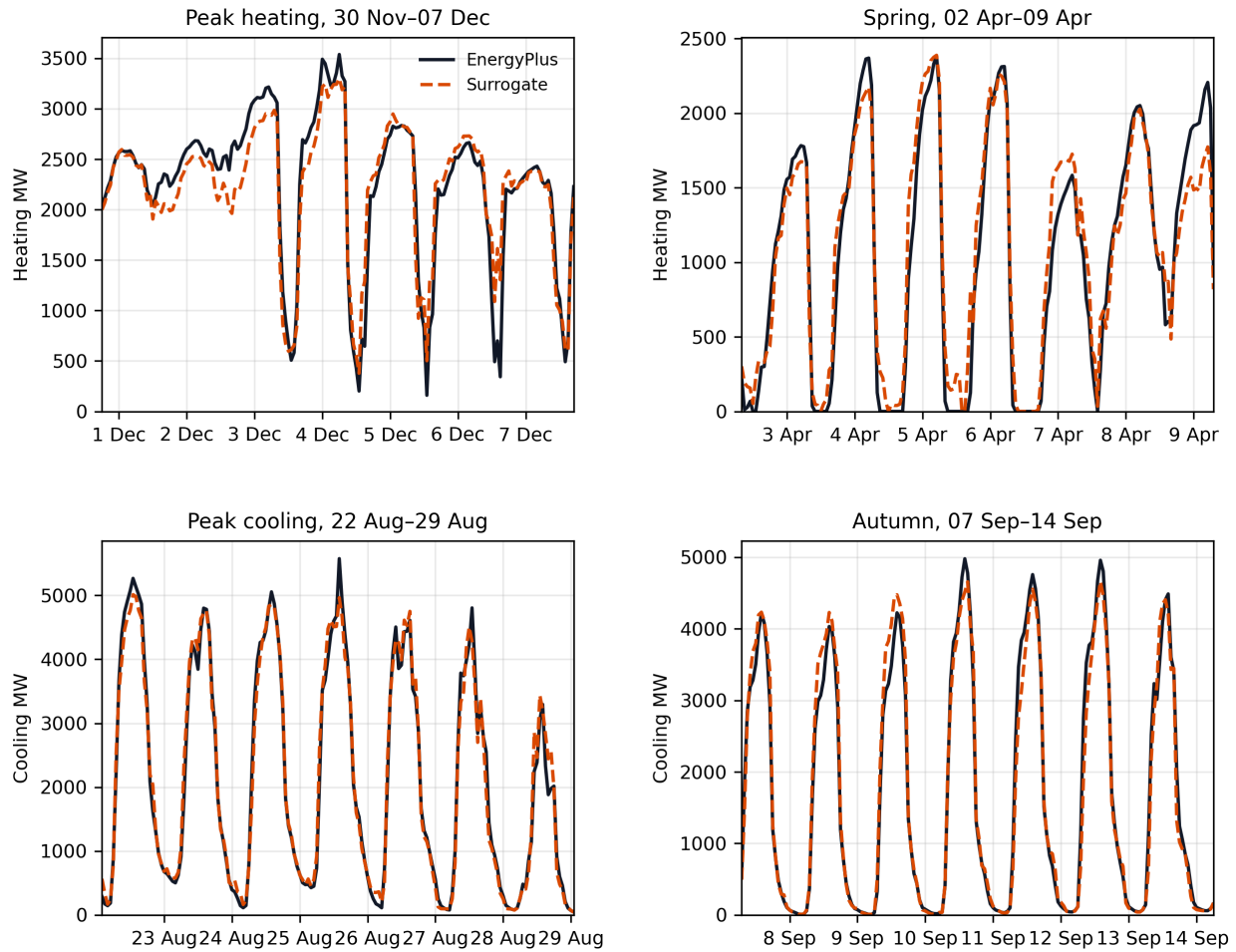


Figure 2: Floor-area-weighted Vienna useful demand from EnergyPlus and held-out surrogate predictions. The panels show the peak-heating week, an April heating week, the peak-cooling week, and a September cooling week. Each interval contains 168 consecutive hours.

After aggregation to the Vienna stock, annual errors remain below 2%, while peak loads are underestimated by 7–10%. Surrogate inference is approximately 39 times faster than the complete EnergyPlus demand path. The resulting model provides a computationally efficient source of EnergyPlus-consistent demand profiles for subsequent energy-system analyses, provided that annual energy anchors and peak-load limitations remain explicit.

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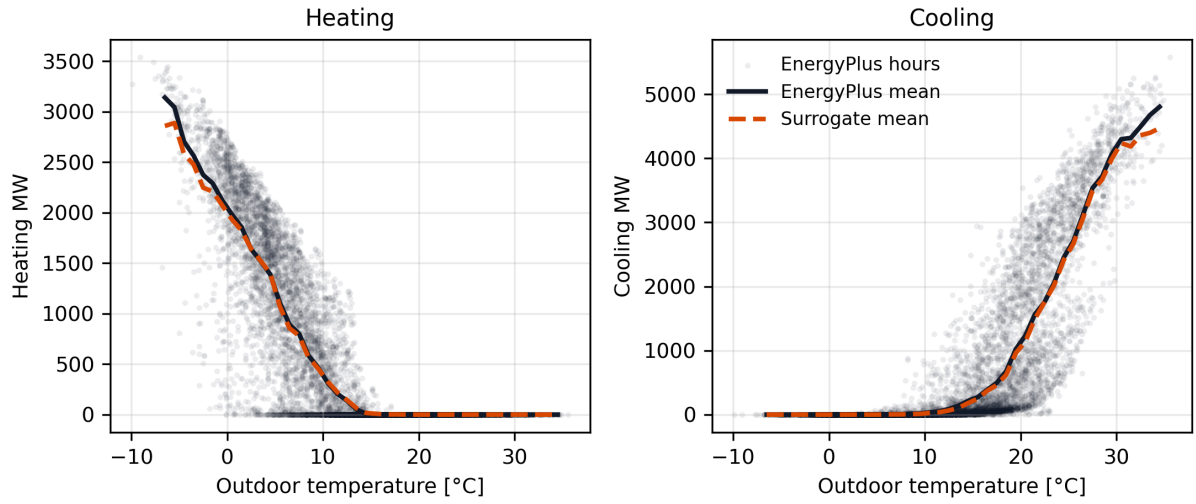


Figure 3: Floor-area-weighted Vienna heating and cooling demand as a function of outdoor temperature. Points show hourly EnergyPlus values. Lines show EnergyPlus and surrogate means in 1-K temperature bins with at least eight observations.

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